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Sébastien **Gradit**

Computational Biologist · Machine Learning for Genomics

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Computational biologist (PhD, Institut Pasteur) building machine-learning and statistical methods for genomics and multi-omics data. First author of **Hicberg** (*Genome Research*, 2026), a framework that recovers chromatin-contact signal from repeated genomic elements that standard pipelines discard. 5+ years across research, clinical metabolomics and medical imaging, shipping reproducible tools, now extending to foundation models for protein and single-cell data.

Core Expertise

ML & AI	**Deep Learning** [*PyTorch, TensorFlow, scikit-learn*], **Foundation Models** [*ESM-2, ESM-C, scGPT, GenePT, ESMFold*]
Omics	**3D Genomics** [*Hi-C*], **NGS** [*WGS*], **LC-MS Metabolomics**, **Multi-Omics Integration**, **Imaging**
Statistics	**Statistical Inference**, **Multivariate Analysis**, **Batch Correction**, **Biomarker Discovery**, **Interpretable ML**
Engineering	**Python**, **R**, **C++**, **SQL**, **Snakemake**, **Docker/Singularity**, **Slurm**, **AWS**, **Git/GitHub Actions**, **uv/Pixi**, **Streamlit**
Languages	**French** [*Native*], **English** [*Fluent, TOEIC 955/990*]

Experience

UVSQ – Université de Versailles Saint-Quentin-en-Yvelines (IHU SEPSIS) — *Versailles, France*
Data Scientist – Clinical Metabolomics & Machine Learning — *2026*

- Built signal-processing and batch-correction pipelines for LC-MS metabolomics on **1,000+ clinical samples**.
- Applied multivariate statistical learning to **5,000+ features** to find sepsis diagnostic signatures.
- Refactored legacy code into modular, tested, FAIR-compliant workflows, cutting processing time by **15%**.

Independent R&D — *Paris, France*
Machine Learning & Bioinformatics Researcher — *2025*

- Co-developed **SynBioGuard** (team), an interpretable **ESM-2**-based classifier flagging biosecurity risks in synthetic-biology designs; **1st place**, Read Write Grow.

Institut Pasteur – Spatial Regulation of Genomes lab — *Paris, France*
PhD Researcher – Computational Genomics & Machine Learning — *2021 – 2024*

- Designed and led **Hicberg**, an open-source statistical/ML framework reassigning multi-mapping reads in repeated genomic elements; first-author paper in ***Genome Research*** (2026), on PyPI/Bioconda.
- Delivered a **30% information gain** in repetitive regions over existing methods with **95% fidelity** to ground truth; generalised to other paired-end omics assays.
- Packaged it with Snakemake, Docker and Slurm workflows so every result re-runs end to end on HPC.

Siemens HealthCare – LIPADE — *Paris, France*
Biomedical Engineer – Machine Learning for Medical Imaging — *2020*

- Built a C++ radiomics engine and ML classifier for bone lesions in xSPECT imaging, speeding up diagnosis by **25%**.
- Developed an explainable segmentation pipeline replacing subjective manual contouring for radiologists.

Danone Nutricia Research — *Saclay, France*
Biomedical Engineer – Clinical Data Science — *2019*

- Built ML-driven bioequivalence simulations and automated R/Python pipelines for clinical trials, accelerating pre-results analysis by **50%** and identifying optimal sample sizes.

Education

PhD in Bioinformatics, Pasteur Institute — *Paris, France* — *2021 - 2024*

M.Sc. in Image Processing and Artificial Intelligence - GPA: 3.33/4.0, Télécom Paris - Sorbonne University — *Paris, France* — *2019 - 2020*

M.Sc. in Biomedical Engineering (Engineer Diploma.) - GPA: 4.0/4.0, EPISEN ISBS - ESIEE — *Paris, France* — *2016 - 2019*

Publications & Software

Hicberg: Reconstruction of Contact Signals from Repeated Elements, **Genome Research** - *2026*
Gradit S., *Ortion S.*, *Larrous P.*, *Delouis M.*, *Koszul R.*, *Cournac A.* - doi: 10.1101/gr.281770.125

Using Statistical Profiling to Decipher Hidden Chromatin Contacts Resulting from Repeated Sequences, **PhD Thesis** - *2024*
Gradit S. - Institut Pasteur

Software: **Hicberg** (sole developer) - distributed on **PyPI** and **Bioconda**, with online documentation, Docker image, Snakemake pipeline and Codecov-tracked tests.

Selected AI Projects

cell-iAI – Predicting Transcriptomic Responses to Genetic Perturbations *D4Gen Challenge*
Team of 3 – github.com/sebgra/cell-iAI *2026*

- Predicted transcriptome-wide expression shifts of unseen gene perturbations (Perturb-seq) from foundation-model gene embeddings (**ESM-2, scGPT, GenePT**), fused by a **mixture-of-experts** model and served through a CLI and API.

SmartAMR – Phage Selection Against Antimicrobial Resistance
Team of 4 – github.com/sebgra/SmartAMR *2025*

- Predicted phage–host relationships from **ESM-C** protein-language-model embeddings of phage and bacterial proteomes (INPHARED, BV-BRC), with binary and multi-label classifiers.

Intelligent Antibodies – Generative In-Silico Antibody Design
Team of 5 – github.com/sebgra/intelligent_antibodies *2024*

- Generated candidate antibody sequences with a **convolutional VAE** and scored antigen binding with a **Siamese CNN–GRU** classifier trained on SAbDab, with **ESMFold** structure prediction in a Streamlit dashboard.

Scientific Communications

2024 **JOBIM 2024, Toulouse**, Poster | Prediction of Omics Signal from Repeated Elements

2024 **JeBiF@JOBIM 2024, Toulouse**, Workshop | Good Practices in Bioinformatics

2023 **International Congress for Transposable Elements, St-Malo**, Poster | Prediction of Omics Signal from Repeated Elements

2023 **31st Intelligence Systems for Molecular Biology (ISMB), Lyon**, Poster and Talk | Statistical Inference of Repeated Elements Contacts in Hi-C maps

Honors & Awards

2025 **1st Place**, Read Write Grow - Biocatalyst Foundation *Europe, Remote*
Team project SynBioGuard: biosecurity screening of synthetic-biology designs combining ESM-2 embeddings, k-mer similarity and GO-term prediction.

2024 **INCEPTION Program Grant**, Fellowship *Paris, France*
Competitive grant funding the final months of the PhD.

2022 **1st Place**, Digital 4 Genomics Hackathon *Genopole, Evry*

Certificates

2023 **TensorFlow Advanced Specialization**, DeepLearning.AI

2023 **Deep Learning Specialization**, DeepLearning.AI

2023 **Machine Learning Specialization**, Stanford - DeepLearning.AI

Leadership & Community

Jeunes BioInformaticiens de France (JeBiF) *France*
President *2022 - 2024*

- Led France's national association of early-career bioinformaticians, running events and training outreach for students and junior researchers.
- Built partnerships with academic labs, companies and international societies, and delivered the JeBiF@JOBIM 2024 workshop on bioinformatics good practices.